Explicit polarized quotients and direct n=5 modular relations

Round 04, 2026-10-18. Parent f5b90fb286dde2e8b0fa15b127a37514a8d64258. The requested geometry and modular-map tracks now have explicit additional constructions. This round evaluates a modular twisting correspondence on closed divisors and collects principal relations on the modular curve. Its relation matrix has rank 4 on 544 columns; no scalar recovery or timing experiment was attempted. The user assesses implications for IC.

Koblitz constructions · round 04

Actual quotient and modular-divisor evaluations · no timing measurements

A5(E_a) × E_a ⟶ Res(E_a) over F2 · q(a,Q)=a−diag(Q)

Natural determinant-5 quotient: equations, rational inverse, 25-point kernels

14 integral quotient modules · both determinant-19 forms have global minimum 3

Other individual graph equations and polarized decomposition remain OPEN

C = X_H(5929), genus 2,841 ⟶ A5(E_a)(F2), order 11

24 supported F64 base inputs · twelve actual Hecke targets each · T11=[4]

Paired bases + canonical Frobenius: 15,840 geometric points ⟶ 544 closed columns

Finite forward divisor projection; global geometric bridges / reverse transfer remain partial

Nonzero projected columns (same population: 544 modular closed points)



Closed degrees: 20 × degree 6; 524 × degree 30. Both signs use F32 coordinate descent.

102 full-fiber principal modular rows · rank 4 / 544 modulo 11

Four closed base fibers force rank ≤ 4; 539 additional ranks to unique scale

Level function x(P)+1: zero degree 9,240; none represented ⟶ row REJECTED

Separate checker: 1,088 point images, 102 rows, BigInt rank, kernels and Hecke sums

11 new named Lean theorems · kernel + axiom audit PASS

Next: nonnatural quotient equations · reverse map / saturation · missing interior divisors

Sources: ledger.json, PROOF_MAP.md, runs.tar.xz. Full scheme/Picard formalization partial.

No individual descent or scalar recovery attempted; user assesses IC implications.

Recorded outcomes

Experiment	Result	Scope limit
Q1	14 explicit quotient bases, integral principal alternating forms and Frobenius matrices	Prime-to-2 polarization presentations; equations for individual nonnatural quotients remain missing
Q1 short vectors	Both distinct determinant-19 Hermitian forms have global minimum 3, with ten vectors of norm 3 each	Minimum on A5, not an indecomposability theorem for the glued fivefold
Q2	Natural quotient A5(E) x E -> Res(E), affine F2 equations and rational inverse for both signs	Target is a principally polarized fivefold; mixed projective charts use the abstract Weil-restriction gluing
Q2 kernel	All 25 geometric kernel points per sign over F_(2^24); 2,420 F2 rational additions checked	No curve/Jacobian realization follows
M1/M2	Actual X0(49) coordinates, two degeneracy maps and degree-11 j-correspondence through extension degree 10	A compatible j-pair alone is not transported level structure
M3b	24 supported F64 base inputs; twelve actual transported Hecke targets each; T11=[4]; fixed Weil pairing	18 ordinary nonzero-j inputs outside the selected trace classes remain unsupported; six j=0 inputs are outside this ordinary chart
M4	15,840 geometric modular points, 544 closed columns: 20 of degree 6 and 524 of degree 30	Selected finite divisor domain; no arbitrary Jacobian normal-form arithmetic

Experiment	Result	Scope limit
M4 images	510 nonzero columns for a=1, 520 for a=0, in their order-11 trace-zero groups	Forward factor projection, not a reverse subgroup embedding into the modular Jacobian
M4 relations	102 principal-function rows, rank 4 modulo 11	539 further independent ranks are needed for unique scale; no individual descent
M5	x(P)+1 has ordinary zero degree 9,240, with zero zeros represented in this factor base	Rejected collector extension; unresolved zeros and poles, no admitted row
Lean	Eleven new named structural theorems build and pass the transitive axiom audit	Modular moduli/descent and full divisor/polarization geometry remain written bridges

The ledger, protocol and amendments, proof map, validation and replay expose the inputs, exact scopes and remaining obligations. All research arithmetic and assembly are Rust. Python only renders this report and the recorded figure. Raw outputs, failed runs and the published modular-polynomial input are frozen in runs.tar.xz.

Explicit polarized quotient geometry

For every retained graph, write its integer top block as G=K*M and set B=[[ell*I8,G],[0,I2]]. The quotient lattice has basis B/ell. The inclusion matrix is Q=[[I8,-G],[0,ell*I2]]. Native exact checks give integral J'=B^t*diag(J_A,ell*J_E)*B/ell^2, determinant one, integral quotient Frobenius, and both polarization-pullback and Frobenius identities. The separate checker repeats these checks with BigInt matrices. The elliptic Frobenius must be its signed integer companion, not positive modular residues; the first failed run exposed that distinction.

The determinant-19 short-vector result is now global. For the doubled real Gram matrix S, every norm-at-most-three vector satisfies z_i^2 <= 6*(S inverse)_ii. Exact inverse cofactors force every coordinate into [-1,1]. Rational LDL enumeration gives ten norm-three vectors and no norm-one or norm-two vectors for each distinct form. A separate exhaustive box check, positive-minor checks and BigInt cofactors agree. The earlier round's bounded statement remains unchanged as historical evidence.

The natural determinant-five graph has actual quotient equations: q(a,Q)=a-diag(Q) maps A5(E) x E onto R=Res_(F32/F2)(E). Its kernel is (diag(T),T), T in E[5]. Over the algebraic closure the target is E^5 with its product principal polarization. Restriction to the augmentation lattice gives H=I4+ones4, while the diagonal block is 5 and mixed terms vanish. The report's universal pairing identity is proved in Lean.

In the pinned F32 basis, expand X=sum(X_i*z^i), Y=sum(Y_i*z^i) and the elliptic equation. Q2 records the five resulting F2 polynomials in ten variables. It checks every affine solution (43 and 21), all rational points (44 and 22), all additions, and all eleven subgroup points. Infinity uses its identity chart. The full Weil-restriction scheme supplies mixed charts; a single affine patch is not represented as a complete projective embedding. The inverse uses the trace to remove the diagonal component. Full five-torsion and all 1,250 kernel additions are checked again with the maintained binary library over F_(2^24).

This supplies one explicit polarized quotient for each sign. The other individual graph quotients, especially the determinant-19 cases with supersingular elliptic cofactor, still have integral polarization data rather than explicit coordinate equations. A complex uniformization for an ordinary variety cannot simply be asserted for those mixed products.

Actual modular transport and factor projections

The modular base is E1=X0(49)/F2: Y^2+XY=X^3+X^2+1. The integral change from the minimal model to eta coordinates reduces to x=X, y=(X^2+X+1)*Y+X+1. This avoids reducing a formula with denominator 2. Its equation and both characteristic-zero and characteristic-two polynomial identities are checked in Lean. The degeneracy parameters are x^4/y and y. The two maps to j are (t^2+t+1)^4/t^7 and (t^2+t+1)^4/t. These come from Kilford--McMurdy, section 2.

M3 transports the two successive cyclic-seven kernels across all twelve cyclic-eleven isogenies, using actual kernel polynomials and characteristic-two Velu x-maps. It checks the commuting square, the published Sutherland degree-11 polynomial, reconstruction on X0(49), and the complete T11=[4] sum. The paired extension uses Miller functions, alternation and a bilinearity control to put every basis at the same primitive eleventh root. The unpaired output is preserved and is not used for relations.

The modular source C is the conjugated full-eleven-level model over X0(49), with canonical determinant-normalized descent. Its analytic group is conjugate to X_H(5929), d congruent to +/-1 modulo 11, of genus 2,841. The explicit moduli/descent argument is in the proof map. Enumerating SL2(F11)/{+/-I} gives all 660 level bases per supported ordinary base point. Canonical Frobenius, including the quadratic-twist correction of the universal elliptic model, produces the 544 closed-point columns. Every one of the 15,840 geometric inputs satisfies the computed twisting-tuple equivariance.

Even sums of Hecke images give the order-five character; signed odd differences give the order-ten character. Integral augmentation and the twisted Frobenius trace evaluate their images in A5(F2). The odd character uses the quadratic isomorphism over F4 before descending to E0(F32). The map outputs actual points, not just Frobenius-module labels. The Gauss-sum identification with the twisted newform is a written geometric bridge using Atkin--Li twisting, not a new full Lean modularity theorem. The finite evaluator supports the recorded closed divisors; arbitrary Jacobian-class reduction, the reverse map and saturation at 11 remain separate requirements.

Direct modular relations and the observed failure

For lines and verticals on X0(49), take their norm from F64 to F2 and pull back the entire divisor to C. Every admitted ordinary fiber includes all 660 level points. Partial rational fibers are rejected. Modular cusps map to cusp points on X0(49), killed by 14, so their

projected order-11 images vanish; the coprime-torsion implication is checked in Lean. Multiplicities, functions and full rows are retained, not inferred from a zero point sum.

The 102 resulting rows are verified against both sets of actual projected points. A separate checker reconstructs every row, repeats its group sum with the maintained binary library, and computes rank 4 with BigInt forward elimination. Their dependence on only four closed base fibers explains the rank ceiling, now proved as a general matrix theorem in Lean. This collector cannot reach rank 543 by collecting more functions with the same four-fiber support pattern.

The next function, x(P)+1, depends on the eleven-level section and could break that pattern. Its exact division polynomial at x=1 has degree 15 in b, with no roots in F64. Its complete ordinary zero degree is 15*56*11=9,240; the current factor base contains none. This attempt therefore admits no row. The next discriminating work is to factor that b-polynomial, construct the missing cyclic49 and complementary-eleven orbits, and resolve the poles. A vanishing image of a partial divisor would not repair the missing support.

Validation, failures and outstanding work

The separate checker verifies 14 quotient modules, two global short-vector certificates, 1,250 geometric-kernel additions, 24 Hecke/basis controls, 1,088 projected point classes, all 544 closed orbits, all 102 principal rows and independent rank 4. Native tests and the full repository library pass. Seven frozen output files replay byte for byte. Eleven named Lean theorems build and audit with standard foundations only. The PDF includes the full proof map and is inspected page by page.

Failed receipts include an incorrect integer lift, the degree-one Rabin edge case, a projective-point counting error in the first Hecke check, first Lean elaborations, a ledger-builder borrow error and a replay mismatch from newly added unpaired metadata. Corrections preserve earlier outputs; matching historical serialization is restored. The rank deficiency and incomplete interior-function support are mathematical/collector outcomes, not build failures or nonexistence claims.

The open requirements are explicit equations/decomposition for nonnatural quotients, full formalization of the modular/Picard bridges, a reverse order-11 subgroup transfer or optimal saturation certificate, richer direct relations and individual descent. No recovery or speed conclusion follows from the current rank. The shared problem remains open. Canonical carrier summaries are updated; performance charts have no new measured quantity to add. Named next experiments are not background jobs.

Formal proof map and written arguments

The pinned Lean 4.34.1/mathlib project now includes KoblitzRound04.lean. Eleven new named theorems build and pass the transitive axiom audit. They use only propext, Classical.choice and Quot.sound, or a subset. Failed proof elaborations are archived; they are not claimed as proofs. No sorry, custom research axiom or trusted native evaluation is used in the final module.

Report claim	Lean theorem	Exact scope and missing bridge
Integral X0(49) coordinate change	modular49_integral_change	Universal commutative-ring identity; identifying the modular eta functions uses the published model
Characteristic-two specialization	modular49_binary_change	Exact identity over every commutative ring of characteristic 2; rational-map chart/gluing remains geometric
Natural quotient polarization pullback	natural_polarization_pullback	Universal bilinear polynomial identity with the fifth coordinate equal to minus the other four; Picard/polarization identification external
Natural quotient kernel	natural_quotient_kernel	All elements of every additive commutative group, trace-zero five-tuple; kernel is diagonal five-torsion
Rational inverse	natural_quotient_rational_inverse	Additive trace/diagonal maps with trace*diagonal=id; on F2 points this follows from five acting as one on E(F2)
Integral augmentation has zero sum	augmentation_sum_zero	Every five-tuple in every additive commutative group
Six-term Frobenius trace is fixed	trace6_fixed	Additive endomorphism and actual period-six input; identifying the twisted modular Frobenius is a separate bridge
Inverse-Gram coordinate bound	dual_coordinate_bound	A real quadratic expression nonnegative for every real t, positive dual norm; positive Gram/inverse specialization supplied by exact certificates and the written argument
Quadratic-twist substitution	binary_twist_substitution	Characteristic 2, s squared+s=1, arbitrary x,y,a,b; supplies the actual curve-equation substitution
Cuspidal projected images vanish	coprime_cusp_image_zero	Every group element killed by both 14 and 11; geometric cusp-image orders are the external premise

Report claim	Lean theorem	Exact scope and missing bridge
Four-fiber rank ceiling	fiber_relation_rank_bound	Every field and every finite matrix factoring through four columns; no empirical rank hypothesis is assumed

Quotient basis, positivity and the explicit natural case

For the first graph of each nonempty gluing case let G=K*M. The subgroup has coordinates (G*z,z)/ell in the product Tate module. Its full inverse image is the lattice with basis B/ell, B=[[ell*I,G],[0,I]]. This basis contains both the old lattice and the graph, and its index is ell squared. The old lattice inclusion is Q=[[I,-G],[0,ell*I]]. Thus Q^t*J'*Q=J_total, where J'=B^t*J_total*B/ell squared. Kernel isotropy supplies its integrality. Its determinant is one, and Frobenius is B inverse*F_total*B, checked integral using F_A*G-G*F_E divisible by ell. These identities, including F'^t*J'*F'=2J', are exact native certificates, not full Lean matrix proofs. For supersingular cofactors this presentation is prime-to-two Tate data; it is not asserted to be an ordinary complex-lattice realization.

For positive real S and a vector z, apply positivity to z-t*S inverse*e_i. It gives 0 <= z^t*S*z-2*t*z_i+t squared*(S inverse)_ii. The generic Lean quadratic lemma implies z_i squared <= (z^t*S*z)*(S inverse)_ii. Here S is twice the real Hermitian Gram, so norm at most 3 gives the factor 6. BigInt determinant/cofactor and positive-minor checks certify the bounds. Exact rational LDL search and a separate complete box enumeration agree: each of the two forms has ten norm-three vectors and none of smaller positive norm. These computations prove the stated minimum with the written positivity-to-lattice bridge; no indecomposability of the final fivefold is inferred.

For the natural graph let A be the kernel of sum:E^5->E after geometric base change, with its F2 permutation descent. The map q:A x E->E^5, (a,Q)->a-diag(Q), is surjective as an isogeny, and its kernel is (diag(T),T), 5T=0. The five-torsion is etale of order 25 in characteristic2. The abstract additive kernel statement is proved in Lean; the isogeny and etale assertions use standard abelian-variety geometry. The product principal polarization pulls back to H=I4+ones4 on A and [5] on E; the mixed terms cancel since sum(a_i)=0. Its degree is 25 squared, so descent through the kernel gives the product principal polarization on the quotient. The target's F2 form is R=Res_(F32/F2)(E), with the descended product polarization. This describes the complete quotient geometrically, plus an explicit affine coordinate patch and identity chart. It is not a fully expanded projective ideal for every mixed chart.

Integral modular model and actual isogeny transport

Kilford--McMurdy, section2 gives the eta model and degeneracy parameters. Starting from Y^2+XY=X^3-X^2-2X-1 over the integers, put x=X-2, D=x^2+7x+7 and y=D*Y+4x^3+22x^2+35x+7. The universal identity in Lean has the eta residual equal to D squared times the Weierstrass residual. Both sides have integral coefficients. Specializing at 2 gives the explicit binary eta map without dividing by 2. D=0 is an exceptional inverse chart, not a discarded point; the base enumeration retains these points and the Fricke involution is evaluated by elliptic addition there.

For ordinary underlying j, normalize the elliptic model to v^2+u*v=u^3+a*u^2+b, b=1/j. Its x-coordinate doubling formula is u squared+b/u squared. A cyclic odd-order kernel with pair-root polynomial k has characteristic-two Velu x-map u+u*k'/k+u squared*(k'/k) squared. Writing T for the sum of the pair roots, its quotient has a4=T, a6=b+T; translating v by T normalizes a4 to zero and changes b to b+T+T squared. The integer-lift certificate records the residual multiple of 2. This Velu construction and Miller's divisor algorithm are standard geometric bridges; the present Lean module does not formalize them.

The code reconstructs both degree-seven kernels from the three j-values of an X0(49) point. Division-polynomial divisibility, separability and doubling invariance check each kernel. Applying the eleven x-map to these kernels is implemented as the characteristic polynomial in their quotient algebras. The transported eleven kernel on the first seven quotient gives an independent commuting-square equality. The native published-polynomial test, curve equation and complete Hecke sum are additional finite controls. The separate maintained library checks basis orders/independence, every transported target, and T11=[4] for all 24 supported inputs.

Modular source, canonical Frobenius and the twisted factor

Let C be the determinant-normalized full-eleven-level cover of X0(49). A point over a field containing mu11 is represented by an ordinary base point with its cyclic49 subgroup and an eleven-torsion basis (P,Q) of fixed Weil pairing zeta, modulo simultaneous sign. There are |SL2(F11)|/2=660 such bases at each ordinary nonelliptic base point.

Conjugation by diag(11,1) takes [[a,b],[c,d]] with c divisible by 49*121 and d=+/-1 modulo 11 to [[a,11*b],[c/11,d]]. The latter group is Gamma0(49) intersect +/-Gamma(11). On moduli the conjugation quotients by the first cyclic11 subgroup and transports the coprime49 structure. The quotient subgroup and the dual generator give the full basis, with the determinant recording its Weil pairing. This supplies the correspondence with the X_H(5929) model; the normalization of the determinant specifies its F2 descent, not just its complex analytic group. This scheme/moduli argument is written, not Lean.

Under F2 Frobenius, canonical normalization is (P,Q)->(P^2,Q^2/2), where division by 2 means multiplication by 6 modulo 11. The pairing transforms as zeta->zeta squared and is restored by that factor. Changes between the chosen universal elliptic twists are explicit y-shifts. The actual permutation of all 15,840 supported geometric points has20 orbits of length6 and 524 of length30. No geometric point is omitted from these selected complete fibers.

Let phi_a be the actual transported X0(49) point associated to the line P+aQ, including the two fixed lines a=0,infinity. For i modulo 5 set D_i^+=phi_(2^i)+phi_(-2^i), and D_i^-=(-1)^i*(phi_(2^i)-phi_(-2^i)). Replace D_i by 5D_i-sum(D_j) to land in augmentation without division. Canonical modular Frobenius transforms the even tuples by sigma followed by inverse cyclic shift, and the odd tuples by its negative. All these equalities are checked at every geometric point, with both trace-zero and order conditions. Their closed-point traces evaluate actual A5(F2) points.

The standard Gauss-sum formula for the weighted degeneracies identifies these with the primitive order-five/order-ten twists of the conductor49 form, up to Galois conjugacy and the explicitly stated integral scaling. Atkin--Li supplies the newform twisting framework. The characteristic-zero correspondence and its prime-to2 level reduction supply the global Picard map. Identification of that map with the optimal modular quotient, its polarization and saturation at 11 are not computed here. The nonzero finite images do not establish a reverse subgroup-to-Jacobian embedding on eleven-torsion.

For the odd sign, choose s in F4 with s squared+s=1 and apply (x,y)->(x,y+s*x). The Lean substitution proves the twist equation. The twisted fixed-point condition then makes both coordinates F32-rational. Their norm to E(F2) vanishes. Since E0(F32) has order 44, E1(F32) has order 22, and the respective base orders are4 and 2, the norm kernels have order 11. The maintained library checks every one of the 1,088 recorded projected points and its eleven-torsion condition. Labels for a planted scalar are never used in relation construction.

Genuine principal rows, cusp removal and the rank ceiling

Take a line or vertical function on E1=X0(49) overF64. Its norm toF2 is a rational function; its full pullback to C has a principal divisor. At each admitted ordinary base zero the entire degree 660 fiber is represented. Base points outside the supported set are rejected, even if a partial projected sum might vanish. The polynomial zero multiplicities determine the row, with coefficients constant on the four supported closed base fibers. The separate verifier reconstructs these rows from their functions and root multiplicities, then checks their point sums using a different binary arithmetic implementation and rank using BigInt elimination.

Every transported cusp is an X0(49) cusp. These are defined overF8 and killed by 14 in E1(F8). Consequently a projected closed cusp class in A5(F2) is killed by 14 and 11 and vanishes; this coprime implication is proved in Lean. A compatible Tate cusp supplies a degree-zero anchor, with zero image. The construction of the anchor and all cusp charts is geometric, not formalized or implemented as arbitrary C divisor-class arithmetic.

Every row factors as W times the four-fiber incidence matrix. The Lean rank lemma proves its rank is at most4 over every field. The computed rank 4 therefore exhausts this row pattern; collecting more of those pullback functions cannot supply the missing539 ranks. A level-dependent function is the next discriminating family, not a supplied cross-row control.

For x(P)+1 the native recurrence gives psi11(1,b) of degree 15, constant coefficient nonzero and no F64 roots. Each ordinary b-root supplies a nonzero point P up to sign,11 complementary paired bases and 56 cyclic49 subgroups. This gives full zero degree 9,240, with multiplicities inherited from the polynomial. None is in the current factor base. Poles also need resolution. The proposed row is rejected rather than substituted for a complete principal divisor. This moduli-degree argument is written; the polynomial and empty-support checks are native exact evidence.

Identity records and remaining formalization

The loose integer identity certificates cover the integral modular change, natural polarization pullback and Velu normalization lift. Their identifiers are listed in ledger.json and cited below after issuance. Each receives 64 issuer points and 128 extra checker points. The first two have direct Lean source bindings. The third retains its multiple-of-two residual and has no misleading Lean attachment.

Full moduli/Picard geometry, polarization descent, Weil/Miller correctness, actual divisor fiber completeness as a scheme theorem, and the reverse projector/composition remain formalization obligations. The finite tests and structural Lean statements are not presented as those full theorems.

Identity certificate	ID	Kernel proof binding
Integral modular change	IDC1h5c93d35226502b13	Direct universal statement, modular49_integral_change
Polarization pullback	IDC1h7a01e3d29f1f37ab	Direct universal statement, natural_polarization_pullback
Velu integer lift	IDC1h5d4af0a173c60eeb	No direct Lean attachment; residual 2*(y*T+T squared) is explicit